

KABILAN RETHINASWAMY

Backend Developer | Python | FastAPI | AWS | Automation & AI

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GitHub: github.com/kabilannadar | LinkedIn: linkedin.com/in/kabilan-rethinaswamy-9738a823a/

Location: [Kanathur, ECR, Chennai, Tamil Nadu](#)

PROFESSIONAL SUMMARY

Backend Developer with hands-on software development experience gained through technical roles and independently developed production-ready projects. Progressively transitioned into backend engineering, developing practical expertise in Python, FastAPI, REST APIs, databases, cloud deployment, automation, real-time systems, and AI-powered applications.

CORE TECHNICAL SKILLS

Programming: Python, Bash, JavaScript, SQL

Backend & APIs: FastAPI, Flask, Node.js, RESTful APIs, WebSockets, SSE, OAuth 2.0, JWT

Databases & ORM: MySQL, PostgreSQL, SQLite, SQLAlchemy

Cloud & DevOps: AWS (EC2, IAM, CloudWatch), Docker, Linux, Git, GitHub, Pytest

AI & ML: RAG, LLMs, Vector Embeddings, Semantic Search, Sentence Transformers, Hugging Face

PROFESSIONAL EXPERIENCE

Technical Support Coordinator

Aurify Systems Pvt. Ltd. | Mar 2025 - Dec 2025

- Developed Python automation and diagnostic utilities for OpenVINO-based client applications, reducing overall manual troubleshooting effort for support teams by approximately 70%.
- Built Python-based failure-detection workflows for DVR/NVR environments, reducing manual troubleshooting effort by approximately 60% through automated failure classification and diagnostics.
- Automated client recovery workflows using Python to detect missed application executions, analyze system events, retry failures, and verify successful recovery when required.
- Reworked database architecture from SQLite to MySQL to eliminate recurring locking issues, while developing automated tests to validate application and automation logic before deployment.

Web Administrator

Sellergize Web Technologies Pvt. Ltd. | Jul 2024 - Nov 2024

- Developed Python-based crawling applications and scripts to retrieve and extract relevant content from multiple provided website sources.
- Built processing logic to clean, normalize, validate, and transform raw crawler output into approximately 80% refined data for downstream PHP workflows.
- Implemented exception handling and logging while continuously testing and refining extraction logic to resolve source-specific crawling and processing issues.
- Supported downstream structured-data workflows through MySQL and phpMyAdmin, working with datasets refined by application and Data teams.

PROJECTS

MoneyCommandAI Assistant - AI-Powered Finance & Budget Assistant

Technologies: Python, FastAPI, Groq (Llama 3.1 8B), RAG, ChromaDB, Sentence Transformers, AI4Bharat IndicTrans2, Socket.IO, JWT

- Built an AI finance assistant that logs expenses from natural language inputs, cutting tedious manual form entries by 60%.
- Developed a ChromaDB vector pipeline using all-MiniLM-L6-v2 embeddings to retrieve relevant financial guidance and prevent LLM hallucinations.
- Built real-time chat using Socket.IO, letting users switch from bot to human agents instantly without losing chat history or repeating themselves.
- Implemented JWT-based role authentication to securely separate access between regular users, support staff, and admin dashboards.

Live Link: expensetrackertn.vercel.app | GitHub: github.com/kabilannadar/MoneyCommandAI-Assistant

ExpenseTracker - Personal Finance & Budget Tracker

Technologies: Python, FastAPI, React.js, PostgreSQL, SQLAlchemy, Alembic, JWT, Google OAuth 2.0, Telegram Bot API

- Unified 12+ specialized modules (Loans, Debts, EMIs, Subscriptions) into one platform, cutting multi-app tracking effort by 80%.
- Integrated MoneyCommandAI Assistant for natural-language financial queries and transaction management, allowing users to log records in under 5 seconds.
- Implemented automated Telegram and email alerts for monthly budgets, EMIs, and subscriptions, ensuring 100% on-time notifications for upcoming dues and limit warnings.
- Built a multi-platform installable PWA (Android, iOS, desktop) featuring offline support with 0% data loss, integrating OAuth 2.0 for secure user access.

Live Link: expensetrackerbykabs.vercel.app | GitHub: github.com/kabilannadar/ExpenseTracker

Automated GitHub-to-JIRA Ticketing System

Technologies: Python, Flask, AWS EC2, IAM, CloudWatch, GitHub Webhooks, JIRA API

- Built a Flask service on AWS EC2 that catches GitHub Webhooks and instantly converts issues into structured JIRA tickets, cutting manual bug-copying effort by 100%.
- Engineered multi-project support so a single GitHub event creates tickets across multiple JIRA boards at once, cutting out duplicate manual entries.
- Integrated AWS CloudWatch and IAM policies using watchtower to record every webhook transaction in real time, giving the team 100% visibility to detect and fix API failures immediately.
- Hosted the app on AWS EC2 with a fixed Elastic IP and security rules, ensuring the webhook URL stays active and never breaks after server restarts.

GitHub: github.com/kabilannadar/flask_jira

EDUCATION & COURSE CERTIFICATIONS

BSc Computer Science

SIES College of Arts, Science and Commerce, Mumbai | 2021 – 2024 | CGPA: 8.28 / 10

100 Days of Code™: The Complete Python Pro Bootcamp - Udemy

57 hours | Completed Apr 2026 | Certificate Link: <https://www.udemy.com/certificate/UC-152f6882-8774-4d15-8651-81349afc0385/>

LANGUAGES

English | Tamil | Hindi | Marathi